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Precision-Guided Munitions: Radar-Guided Weapons



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Radar guidance like infrared (IR) homing guidance is predominantly used in surface-to-air and air-to-air guided weapons. While IR homing guidance is largely employed in short and medium operational ranges, radar guidance is particularly attractive for long-range air-to-air guided missiles. State-of-the-art radar-guided air-to-air missiles using active radar homing guidance technology have operational ranges in excess of 150km. This concluding part of the article focuses on operational aspects of different types of radar-guided weapons. Features and capabilities of major international radar-guided weapon systems are also discussed.

Radar-guided weapons are surface-to-air and air-to-air anti-aircraft-guided missiles in which a radar is used to designate the intended target with electromagnetic energy, and the missiles make use of radar energy reflected from the target to steer itself to intercept and destroy the target. The basic concept is the same as the one used in the case of laser-guided munitions, except for the fact that laser-guided munitions are largely surface-to-surface and air-to-surface weapons and radar-guided weapons are mainly air-to-air and to a limited extent surface-to-air missiles.

Another difference is in their operational ranges, which are much larger in the case of radar-guided missiles, approaching 150km to 200km in the case of state-of-the-art air-to-air missiles than in the case of laser-guided munitions that seldom exceed 20km.

AIM-54 Phoenix and AIM-120D AMRAAM have maximum operational ranges of 190km and 180km, respectively. Also, laser is either on a land based platform or aircraft mounted in semi-active hom-

ing guidance. It is an integral part of the weapon hardware in the case of active laser homing.

In the case of radar-guided weapons, the radar is also mounted either on a land vehicle or an aircraft in semi-active radar homing guidance. Radar is an integral part of the guided-missile hardware in the case of active radar homing guidance.

A variation of semi-active radar guidance is track-via-missile (TVM) guidance. Like semi-active homing missiles, the ground based radar designates the target with radar energy, which is then reflected off the target and detected by the radar receiver onboard the missile.

However, unlike a semi-active radar homing missile, the missile does not have hardware and software to compute track and intercept information. Instead, data from the reflected radar energy is relayed back to the ground station via a data link. The ground station computes the track information and communicates it back to the missile.

Radar-beam-riding guidance is another technique that was used in earlier days for short-range surface-to-surface and surface-to-air missiles. But because of inherent shortcomings of the beam-riding concept, both laser and radar beam-riding weapons are more or less an extinct entity.

Semi-active radar guidance

In a semi-active radar guidance system, the target aircraft is illuminated by electromagnetic energy emitted by the fire control radar located either on the launch aircraft or at an appropriate ground location. The fire control radar acquires the target and tracks it. A small radar transmitter generating a very narrow beam then selectively illuminates the target using tracking information generated by the fire control